

Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data

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UE24CS352A Machine Learning mini project, PES University. Pipeline demonstrated on ABIDE resting-state fMRI

Problem statement

Excessive alcohol use in adolescence changes brain structure, but its effect on functional connectivity is unclear. The task, following Kim, Liu and Noh (Stanford CS229), is binary classification of a person from their resting-state fMRI: brain activity (BOLD signal) recorded at rest, summarised as one time series per brain region. We reproduce the paper's baseline, test stronger connectivity features, and check whether predictions rely on age or on scanner site.

Dataset

The paper used NCANDA (715 adolescents). Access needs an NIAAA Data Use Certification and a committee review of up to 30 days, which does not fit a one-week project. We therefore use ABIDE Preprocessed (CPAC pipeline, band-pass filtered, quality checked), the closest public match: resting-state fMRI, the same Craddock atlas family (200 regions), one time series per region, a multi-site imbalanced sample. It has 871 people (403 autism, 468 control) from 20 sites, 78 to 316 timepoints per scan, aged 6.5 to 58.0. The label is autism vs control instead of heavy drinking. The pipeline is dataset agnostic: using NCANDA only needs a new loader returning the same arrays.

Approach

Features: (1) the paper's derived feature, max minus min of each region's signal (200 values); (2) functional connectivity, the correlation between every pair of regions (19,900 values); (3) tangent-space connectivity, which measures each person relative to a group mean learned on the training fold; (4) demographics: age, sex, one-hot site (the equivalent of the paper's scanner type). Each brain feature set is run alone, with demographics, and demographics alone is run separately (age ablation).

Models: L2 logistic regression, linear SVM and RBF SVM with balanced class weights (all people are kept, no downsampling), a majority-class dummy baseline, and a small 1D CNN on z-scored raw time series (random 64-step crops, global average pooling).

Evaluation: 5-fold stratified CV and 5-fold site-grouped CV (no site appears in both train and test). Anything learned from data (tangent reference, scaler) is fit on training folds only, and C is tuned by an inner 3-fold grid search. We report the mean and standard deviation of accuracy, F1 and ROC AUC. Seed 42 everywhere.

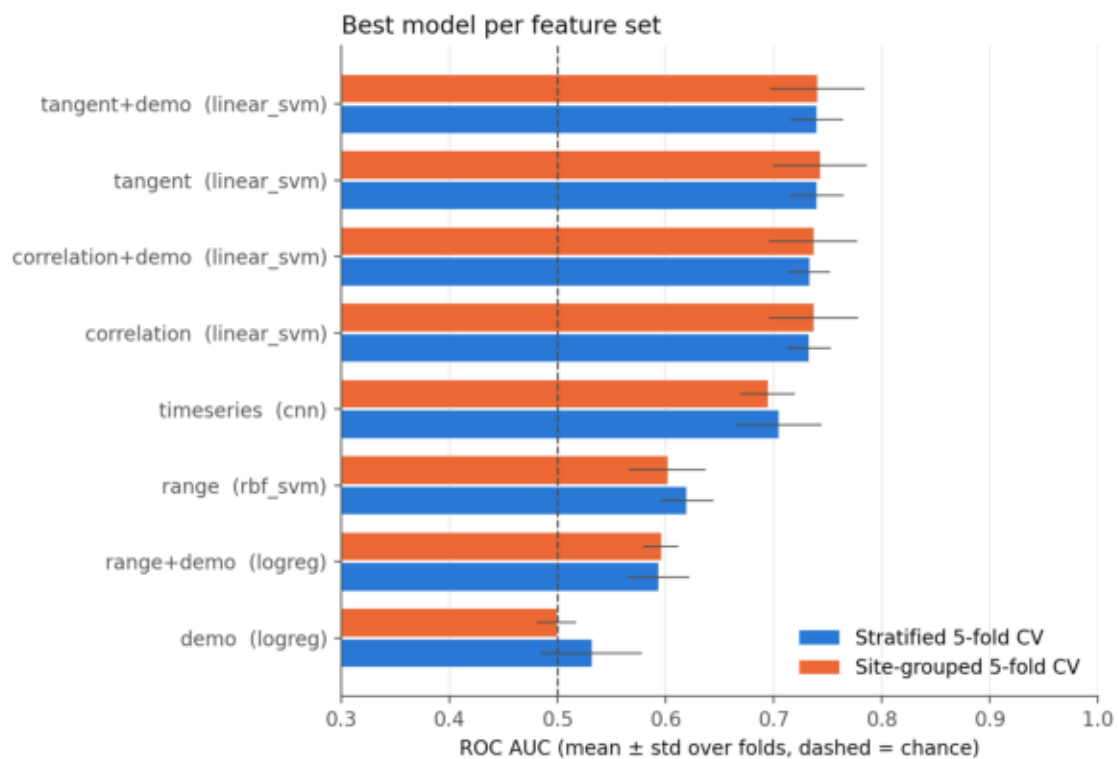
Implementation

Python with nilearn, scikit-learn and PyTorch. `src/data.py` loads data, `src/features.py` builds features, `src/models.py` builds the nested pipelines, `src/evaluate.py` runs CV, `src/deep.py` holds the CNN and `src/plots.py` the figures. `run_experiments.py` writes every number to `results/` and `figures/`; `demo.py` runs offline and predicts people the demo model never saw; `make_report.py` builds this document and the slides from those files.

Results

- Best model: tangent+demo features with Linear SVM, accuracy 0.675 ± 0.021 and ROC AUC 0.740 ± 0.024 (stratified 5-fold CV), against 0.537 accuracy and AUC 0.5 for always guessing the majority class. Out of fold it caught 273/403 autism cases (sensitivity 0.68) and 315/468 controls (specificity 0.67).
- Paper reproduction: range features + demographics with logistic regression gave accuracy 0.563 (AUC 0.594); range features alone gave 0.583 (AUC 0.597). The paper reported 0.713 with age and 0.538 without on NCANDA (different task, so not directly comparable).
- Better features: correlation connectivity beats the paper's range features (best AUC 0.733 vs 0.620).
- Age ablation: demographics alone reach AUC 0.532; tangent brain features alone reach 0.740 and 0.740 with demographics added. Unlike the paper, the prediction is carried by the brain features, not by age.
- Site effect: holding out whole sites moves the best model's AUC from 0.740 to 0.741 (+0.001). The model transfers to unseen scanners.
- Deep learning: the 1D CNN on raw time series reached accuracy 0.634, AUC 0.705, below the linear models on connectivity: as in the paper, simple models on derived features win on small fMRI data.

Features	Model	Acc	F1	AUC	Site acc	Site AUC
demo	Logistic regression	0.532 ± 0.017	0.476	0.532 ± 0.047	0.481	0.499
range	RBF SVM	0.583 ± 0.027	0.547	0.620 ± 0.025	0.569	0.602
range+demo	Logistic regression	0.563 ± 0.022	0.537	0.594 ± 0.029	0.569	0.596
correlation	Linear SVM	0.668 ± 0.017	0.648	0.733 ± 0.020	0.673	0.737
correlation+demo	Linear SVM	0.667 ± 0.017	0.646	0.733 ± 0.020	0.673	0.737
tangent	Linear SVM	0.677 ± 0.023	0.660	0.740 ± 0.025	0.674	0.743
tangent+demo	Linear SVM	0.675 ± 0.021	0.658	0.740 ± 0.024	0.664	0.741
timeseries	1D CNN	0.634 ± 0.028	0.613	0.705 ± 0.040	0.643	0.695
none (dummy)	Dummy (majority)	0.537 ± 0.003	0.000	0.500 ± 0.000	0.537	0.500



Conclusions

- Resting-state connectivity separates the classes above chance (0.675 vs 0.537 accuracy), and site-grouped CV (0.664) is the honest number to quote.
- Whole-brain correlation features are a stronger and simpler starting point than per-region signal range.
- Always report a dummy baseline, a demographics-only model and site-held-out CV: without them a model can look good for the wrong reasons (age or scanner).
- Next step: run the same pipeline on NCANDA once the data use agreement is approved (only src/data.py changes).

Reference: Y. Kim, C. Liu, J. Noh, Classifying Adolescent Excessive Alcohol Drinkers from fMRI Data, CS229, Stanford.